WATER PURIFICATION



R26-IT-005

Progress Presentation 2

SLIIT
FACULTY OF COMPUTING

2026/08/31

Comments for the PP1 and how you/team addressed them

STUDENT'S NAME & ID	COMMENTS FOR THE PP1	HOW YOU/TEAM ADDRESSED THEM
R.M.N.A. RATHNAYAKE IT22193940	• Only have brought the device, but the device also doesn't work. • No software implementation, needs to discuss with supervisor	• Developed a software solution to analyse water quality inputs and provide automated results, addressing the lack of software implementation.
W.N.N. DE SILVA IT22211750	• Have trained only 1 model, better to train 2 models and do a comparison on model performance. • 4 Quality classifications with water samples and test the water	• Trained two machine learning models, compared their performance using evaluation metrics, and selected the better-performing model for water quality classification.
D. DODANWELA IT22212122	• Only have brought the device, but the device also doesn't work. • No software implementation, needs to discuss with supervisor	• Developed the required software component to analyse water quality data and provide automated results.
W.A.S.P.D. WIJESINGHA IT22295460	• Only have brought the device, but the device also doesn't work. • No software implementation, needs to discuss with supervisor	• Implemented the software component for automated water contamination detection and analysis.



Problem

- Unsafe water causes health risks and waterborne diseases.
- Water testing often requires laboratory equipment and trained personnel.
- Existing methods can be costly and time-consuming.
- Many users cannot easily identify contamination in water.



Stakeholders

- Households
- Rural communities
- Water treatment authorities
- Environmental organizations
- Schools and universities
- Researchers

Impact and Measurable Pain Point



Research Gap



Approach	Comprehensive?	Automated?	Unified?
Visual Inspections	No	No	No
Sensor-Based Monitoring	No	Yes	No
Microbial Tests	No	No	No
Isolated AI Models	No	Yes	Yes
Purification Guidance	No	No	No

No existing solution combines contamination detection, parameter analysis & purification advice into a single intelligent system



Objectives

1.Detect visible contaminants in water samples.
2.Classify water quality using measured parameters.
3.Detect microorganisms from microscope images.
4.Generate an automated water safety report.
5.Recommend suitable purification methods.

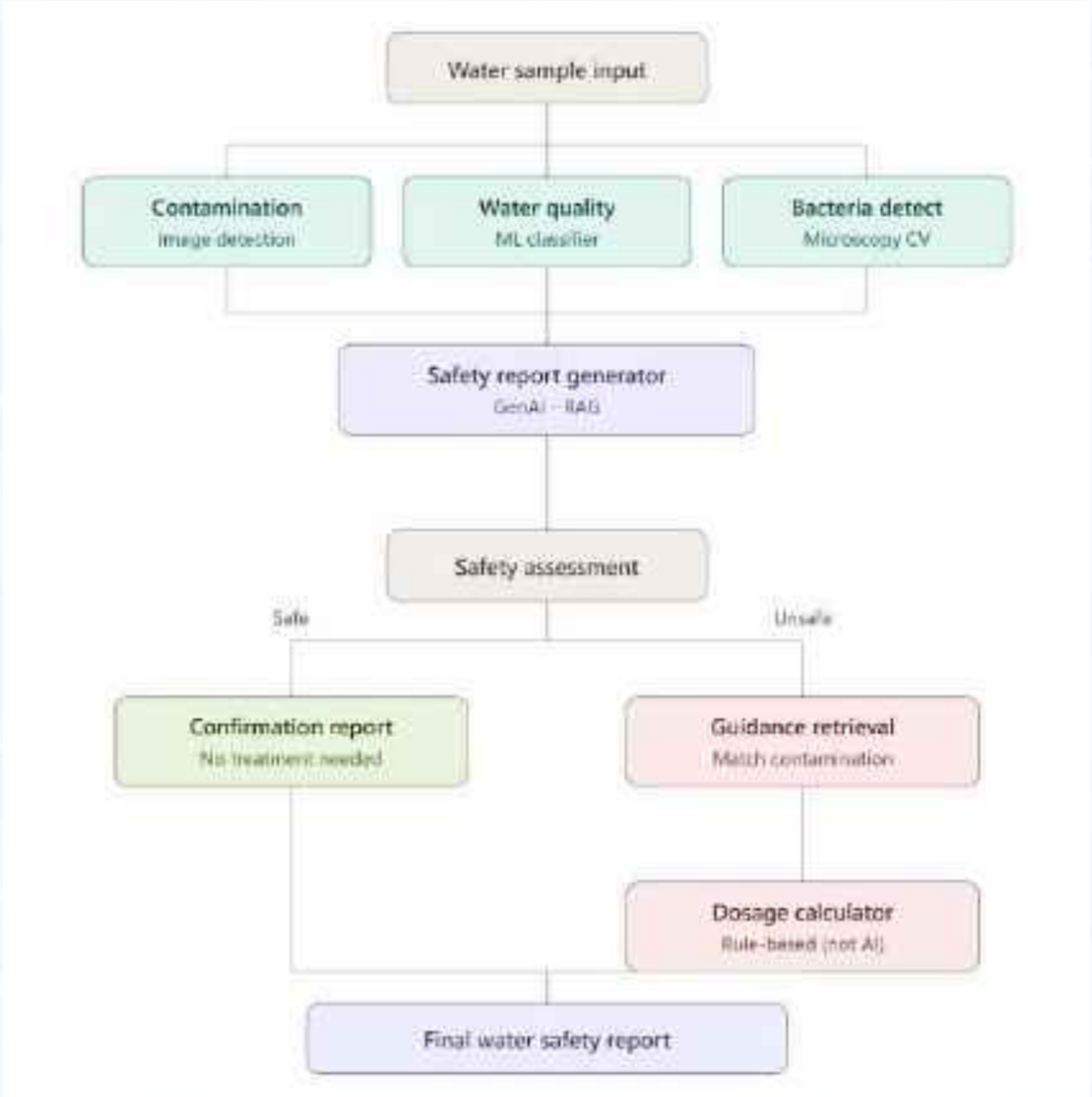


Solution Overview + Architecture



System Design & Delivery Practices:

- **Modular multi-component ecosystem** — independently trained & evaluated models
- **Multi-source data fusion** — visual, sensor, and microbial inputs combined
- **Multi-criteria decision logic** — contamination severity, safety grade, and purification priority
- **API-first, three-tier architecture** — mobile app, dashboard, and backend integration
- **CI/CD pipeline** — automated model training, testing & deployment
- **Explainable-by-design** — transparent contamination reasoning & purification recomendations


Water sample input
Contamination
Image detection
Water quality
ML classifier
Bacteria detect
Microscopy CV
Safety report generator
GenAI – RAG
Safety assessment
Safe
Unsafe
Confirmation report
No treatment needed
Guidance retrieval
Match contamination
Dosage calculator
Rule-based (not AI)
Final water safety report

Integration and Completion Status



Evaluation Evidence

Model	Accuracy	F1	ROC-AUC
Contamination Detection (DL-Based)	91.56%	0.9165	0.9857
Microorganism Detection (DL-Based)	89.32%	0.8937	0.9862
Water Quality Classifier (ML-Based)	93.68%	0.9410	—
Automated I Classifier (ML-Based)	93.68%	0.9410	—

85.22% → 93.68%*

Water Quality Classifier

Accuracy Improvement

App-4-Backend

• Automanes Dapagnnien

25.41% → 1.69%

Detection Error Rate

Over 15x reduction

Non-Functional Requirements



Performance

The system should process water sample images, sensor parameters, and microscopy images within an acceptable response time.

Accuracy

The contamination, water-quality, and microorganism detection models should provide reliable results with high classification / detection accuracy.

Reliability

The system should consistently produce results without unexpected failures during water sample assessment.

Availability

The system should be available whenever users need to perform water quality or contamination assessments.

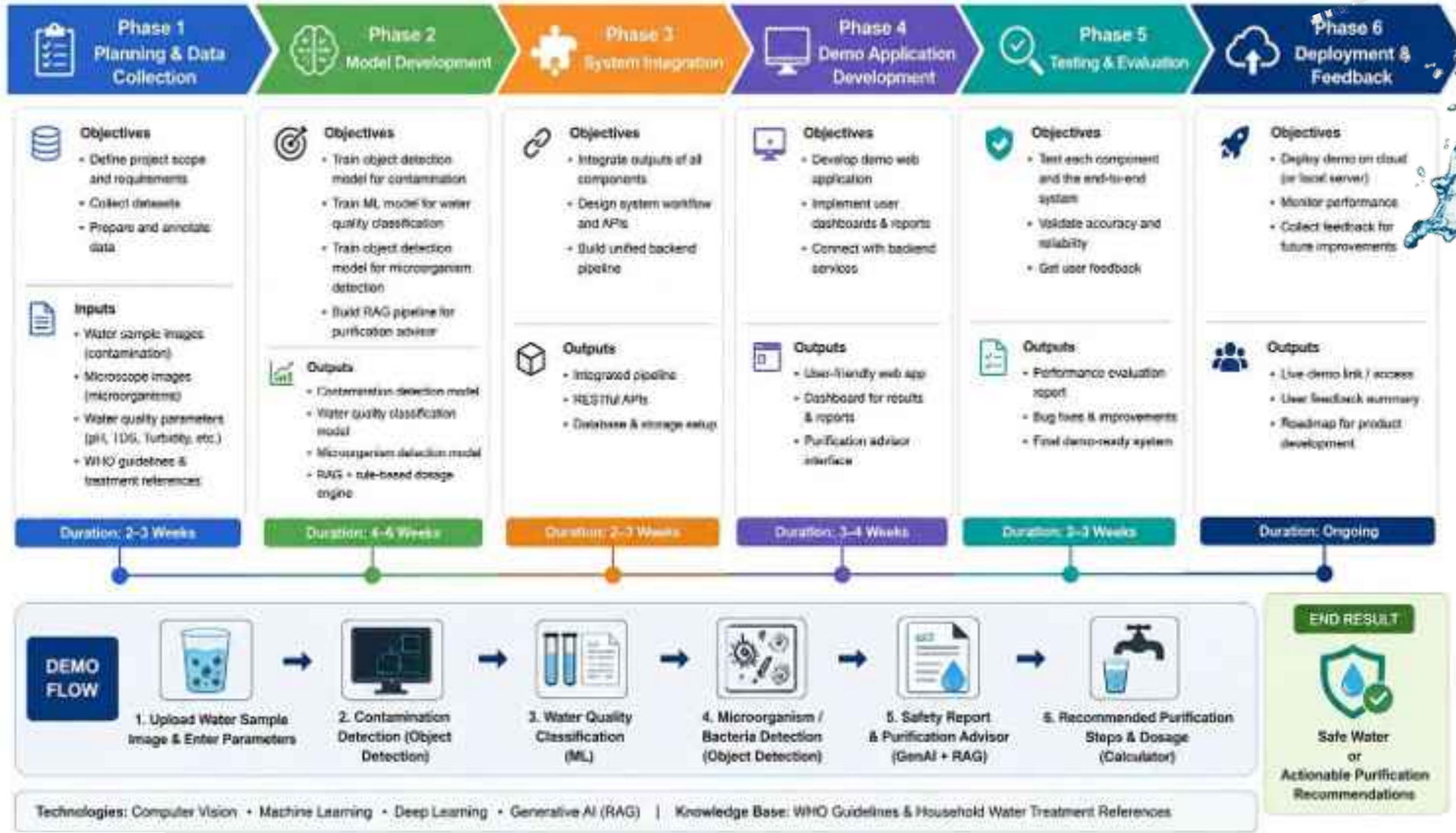
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Commercialization



Target Market	• Water quality testing laboratories and environmental monitoring organizations • Drinking-water suppliers and water treatment companies • Government and public health authorities • Households and communities requiring water safety assessment
Value Proposition	• Detect visible water contamination automatically using computer vision • Classify water quality using physicochemical parameters such as pH, turbidity, TDS and fluoride • Detect microorganisms and bacteria from microscope images • Generate water safety reports and personalized purification recommendations
Investment & Cost Recovery	• Development cost: AI model training, cloud hosting, datasets and system maintenance • Revenue through API / platform subscription services • Licensing the water quality assessment and purification advisory system • Potential integration with existing water-testing and treatment platforms
Intellectual Property	• Deep learning models for visual contamination and microorganism detection • Machine learning-based water quality classification framework • Integrated water safety assessment and purification recommendation system • Software copyright and potential patent protection

Demo Roadmap




Step 1 — Surface Contamination Scan






Step 3 — Drinking Water Chemistry


Step 3 — Drinking Water Chemistry


Step 4 — Water Quality Report

THANK YOU

ANY QUESTIONS ?

